

# Package ‘PhyloTraitDynamics’

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**Title** Companion Code for Phylogenetic Trait Dynamics

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**Description** Conservative companion package for reproducing numerical results and figures for an article on birth-death processes, MRCA ages, and Brownian empirical moments on phylogenies. The package exposes a small public API with package-local implementations.

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grDevices,  
stats

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---

birth\_death\_brownian\_simulate

*Simulate Birth-Death Brownian Empirical Moments*


---

## Description

Simulates birth-death Brownian paths and records, on the requested time grid, the number of living lineages, the empirical mean of their traits, and their empirical variance.

## Usage

```
birth_death_brownian_simulate(
  birth,
  death,
  sigma2 = 1,
  time_end,
  time_step,
  B,
  x0 = 0,
  seed = NULL,
  n_envelope = 4000,
  safety_factor = 1.05
)
```

## Arguments

|                     |                                                                           |
|---------------------|---------------------------------------------------------------------------|
| birth               | Non-negative birth rate, either a numeric constant or a function of time. |
| death               | Non-negative death rate, either a numeric constant or a function of time. |
| sigma2              | Brownian variance parameter.                                              |
| time_end, time_step | Time grid definition.                                                     |
| B                   | Number of simulated paths.                                                |
| x0                  | Initial trait value.                                                      |
| seed                | Optional random seed.                                                     |
| n_envelope          | Number of points used by the thinning envelope.                           |
| safety_factor       | Multiplicative safety factor for the thinning envelope.                   |

## Value

An object of class "birth\_death\_brownian\_simulation" containing the time grid, matrices of empirical means, empirical variances and lineage counts, and a simulation summary.

---

empirical\_mean\_compute\_variance

*Variance of the Empirical Mean Under Birth-Death Brownian Dynamics*


---

### Description

Computes the theoretical variance of the empirical mean of Brownian traits, conditional on survival at each time.

### Usage

```
empirical_mean_compute_variance(
  birth,
  death,
  sigma2 = 1,
  time_end,
  time_step,
  n_steps = 1200,
  tol_sing = 1e-07,
  li2_tol = 1e-12
)
```

### Arguments

|                     |                                                                           |
|---------------------|---------------------------------------------------------------------------|
| birth               | Non-negative birth rate, either a numeric constant or a function of time. |
| death               | Non-negative death rate, either a numeric constant or a function of time. |
| sigma2              | Brownian variance parameter.                                              |
| time_end, time_step | Time grid definition.                                                     |
| n_steps             | Number of integration steps for numerical theory.                         |
| tol_sing, li2_tol   | Numerical tolerances.                                                     |

### Value

A data frame with time, survival probability, empirical mean variance, and conditioning columns. The original script result is attached as attribute "original\_result".

---

empirical\_mean\_plot\_variance

*Plot Variance of the Empirical Mean*


---

### Description

Plots Monte Carlo summaries and/or theoretical values for the variance of the empirical mean.

**Usage**

```
empirical_mean_plot_variance(
  simulation = NULL,
  theory = NULL,
  summary = NULL,
  main = NULL,
  xlab = "time",
  ylab = "Variance of empirical mean"
)
```

**Arguments**

|                  |                                                                                                                                                              |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| simulation       | Optional "empirical_mean_simulation" object.                                                                                                                 |
| theory           | Optional data frame returned by <code>empirical_mean_compute_variance()</code> .                                                                             |
| summary          | Optional data frame returned by <code>empirical_mean_summarise_simulation()</code> .<br>If omitted and simulation is supplied, it is computed automatically. |
| main, xlab, ylab | Base graphics labels.                                                                                                                                        |

**Value**

Invisibly returns a list containing simulation, theory, and summary.

---

empirical\_variance\_compute\_expectation

*Expected Empirical Variance Under Birth-Death Brownian Dynamics*

---

**Description**

Computes the theoretical expectation of the empirical trait variance on a time grid for a birth-death process with Brownian trait evolution.

**Usage**

```
empirical_variance_compute_expectation(
  birth,
  death,
  sigma2 = 1,
  time_end,
  time_step,
  conditioning = c("none", "survival"),
  method = c("auto", "numeric", "closed_constant"),
  n_steps = 2000,
  tol = 1e-08,
  constant_tol = 1e-10,
  li2_tol = 1e-12
)
```

**Arguments**

|                            |                                                                                                                                                                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| birth                      | Non-negative birth rate, either a numeric constant or a function of time.                                                                                                                                                                |
| death                      | Non-negative death rate, either a numeric constant or a function of time.                                                                                                                                                                |
| sigma2                     | Brownian variance parameter.                                                                                                                                                                                                             |
| time_end, time_step        | Time grid definition.                                                                                                                                                                                                                    |
| conditioning               | Conditioning used for the expectation. "none" is unconditional and "survival" conditions on $N(t) > 0$ .                                                                                                                                 |
| method                     | Theory evaluation method. "auto" uses the closed constant-rate formula when the birth and death rates are constant on the time grid, and otherwise uses the numerical method. "closed_constant" requires constant birth and death rates. |
| n_steps                    | Number of integration steps for numerical theory.                                                                                                                                                                                        |
| tol, constant_tol, li2_tol | Numerical tolerances passed to the original implementation.                                                                                                                                                                              |

**Value**

A data frame with time, survival probability, expected empirical variance, conditioning, and method columns. The original script result is attached as attribute "original\_result".

---

empirical\_variance\_plot\_expectation

*Plot Expected Empirical Variance*


---

**Description**

Plots simulated empirical variance paths, Monte Carlo summaries, and/or the theoretical expectation on a common set of axes.

**Usage**

```
empirical_variance_plot_expectation(
  simulation = NULL,
  theory = NULL,
  summary = NULL,
  show_paths = TRUE,
  conditioning = c("none", "survival"),
  max_paths = Inf,
  alpha = 0.05,
  main = NULL,
  xlab = "time",
  ylab = "Empirical variance"
)
```

**Arguments**

|                  |                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| simulation       | Optional "empirical_variance_simulation" object.                                                                                                |
| theory           | Optional data frame returned by empirical_variance_compute_expectation().                                                                       |
| summary          | Optional data frame returned by empirical_variance_summarise_simulation(). If omitted and simulation is supplied, it is computed automatically. |
| show_paths       | Logical; whether to draw individual simulated paths.                                                                                            |
| conditioning     | Which expectation to plot.                                                                                                                      |
| max_paths        | Maximum number of simulated paths to draw.                                                                                                      |
| alpha            | Transparency used for simulated paths.                                                                                                          |
| main, xlab, ylab | Base graphics labels.                                                                                                                           |

**Value**

Invisibly returns a list containing simulation, theory, and summary.

---

fixed\_tree\_compute\_theoretical\_summary

*Theoretical Summary on a Fixed Tree*

---

**Description**

Computes theoretical summaries of empirical mean variance and empirical variance for Brownian traits on a fixed phylogenetic tree.

**Usage**

```
fixed_tree_compute_theoretical_summary(
  tree,
  sigma2 = 1,
  time_end = NULL,
  time_step
)
```

**Arguments**

|           |                                                               |
|-----------|---------------------------------------------------------------|
| tree      | A phylogenetic tree of class "phylo".                         |
| sigma2    | Brownian variance parameter.                                  |
| time_end  | End of the time grid. If NULL, the maximum tree time is used. |
| time_step | Time step used to define the time grid.                       |

**Value**

A data frame with time, lineage counts, empirical mean variance, empirical variance expectation, empirical variance quartiles, and empirical variance variance. The tree, sigma2, and original result are attached as attributes.

---

`fixed_tree_plot_brownian_realization`*Plot a Fixed-Tree Brownian Realization*

---

**Description**

Plots the simulated Brownian realization together with the fixed tree using the original plotting routine.

**Usage**

```
fixed_tree_plot_brownian_realization(  
  x,  
  n_bands = 10,  
  time_band_col = grDevices::adjustcolor("blue", alpha.f = 0.05),  
  cex = 1  
)
```

**Arguments**

|                            |                                                                |
|----------------------------|----------------------------------------------------------------|
| <code>x</code>             | A <code>fixed_tree_brownian_realization</code> object.         |
| <code>n_bands</code>       | Number of time bands shown by the original plotting routine.   |
| <code>time_band_col</code> | Color used for time bands.                                     |
| <code>cex</code>           | Character expansion factor passed to base graphics parameters. |

**Value**

Invisibly returns `x`.

---

`fixed_tree_plot_theoretical_summary`*Plot a Fixed-Tree Theoretical Summary*

---

**Description**

Plots fixed-tree theoretical summaries for empirical mean variance and empirical variance, optionally with the tree.

**Usage**

```
fixed_tree_plot_theoretical_summary(  
  x,  
  tree = NULL,  
  show_tree = TRUE,  
  main = NULL,  
  time_band_col = grDevices::adjustcolor("blue", alpha.f = 0.05),  
  band_col = grDevices::adjustcolor("red", alpha.f = 0.18),  
  expectation_col = "blue",  
  median_col = "red",  
)
```

```

    mean_variance_col = "darkgreen",
    cex = 1
)

```

### Arguments

|                   |                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------|
| x                 | A fixed_tree_theoretical_summary object, as returned by fixed_tree_compute_theoretical_              |
| tree              | Optional phylogenetic tree of class "phylo". If NULL, the tree attached to x is used when available. |
| show_tree         | Logical; whether to show the tree panel.                                                             |
| main              | Optional main title.                                                                                 |
| time_band_col     | Color used for time bands.                                                                           |
| band_col          | Color used for the empirical-variance interquartile band.                                            |
| expectation_col   | Color used for the empirical-variance expectation.                                                   |
| median_col        | Color used for the empirical-variance median and quartiles.                                          |
| mean_variance_col | Color used for the empirical-mean variance.                                                          |
| cex               | Character expansion factor passed to base graphics parameters.                                       |

### Value

Invisibly returns x.

---

fixed\_tree\_simulate\_brownian\_realization

*Simulate Brownian Traits on a Fixed Tree*

---

### Description

Simulates one Brownian trait realization on a fixed phylogenetic tree and records empirical mean and empirical variance through time.

### Usage

```

fixed_tree_simulate_brownian_realization(
  tree,
  sigma2 = 1,
  time_step,
  seed = NULL
)

```

### Arguments

|           |                                           |
|-----------|-------------------------------------------|
| tree      | A phylogenetic tree of class "phylo".     |
| sigma2    | Brownian variance parameter.              |
| time_step | Time step used for the simulation output. |
| seed      | Optional random seed.                     |

**Value**

An object of class "fixed\_tree\_brownian\_realization" containing the input tree, simulation data frame, time series, parameters, and seed.

---

mrca\_age\_compute\_distribution

*MRCA Age Distribution at One Time*


---

**Description**

Computes the distribution and summary statistics of the most recent common ancestor age at a fixed time under a birth-death process.

**Usage**

```
mrca_age_compute_distribution(
  birth,
  death,
  time,
  time_step = NULL,
  probs = c(0.25, 0.5, 0.75),
  n_internal = NULL
)
```

**Arguments**

|            |                                                                                 |
|------------|---------------------------------------------------------------------------------|
| birth      | Non-negative birth rate, either a numeric constant or a function of time.       |
| death      | Non-negative death rate, either a numeric constant or a function of time.       |
| time       | Observation time.                                                               |
| time_step  | Internal time step. If NULL, a default step is derived from time.               |
| probs      | Quantile probabilities to report.                                               |
| n_internal | Optional number of internal subintervals; when supplied it overrides time_step. |

**Value**

An object of class "mrca\_age\_distribution" with distribution grid, summary statistics, requested probabilities, and the original result.

---

mrca\_age\_compute\_dynamics

*MRCA Age Distribution Dynamics*


---

## Description

Computes MRCA age distribution summaries over a time grid under a birth-death process.

## Usage

```
mrca_age_compute_dynamics(
  birth,
  death,
  time_end,
  time_step,
  probs = c(0.25, 0.5, 0.75),
  n_internal_per_step = 10L,
  include_zero = TRUE
)
```

## Arguments

|                     |                                                                             |
|---------------------|-----------------------------------------------------------------------------|
| birth               | Non-negative birth rate, either a numeric constant or a function of time.   |
| death               | Non-negative death rate, either a numeric constant or a function of time.   |
| time_end, time_step | Time grid definition.                                                       |
| probs               | Quantile probabilities to report.                                           |
| n_internal_per_step | Number of internal integration substeps per output time step.               |
| include_zero        | Logical; whether to include time zero in the original dynamics computation. |

## Value

An object of class "mrca\_age\_dynamic" containing the summary data frame, distribution grids, probabilities, and original call.

---

mrca\_age\_plot\_dynamics

*Plot MRCA Age Distribution Dynamics*


---

## Description

Plots MRCA age summaries over time, including median, optional interquartile band, and optional expectation.

**Usage**

```
mrca_age_plot_dynamics(  
  x,  
  show_iqr = TRUE,  
  show_expectation = TRUE,  
  main = NA,  
  xlab = "Time",  
  ylab = "MRCA age",  
  add_legend = TRUE,  
  ...  
)
```

**Arguments**

|                  |                                                                   |
|------------------|-------------------------------------------------------------------|
| x                | An "mrca_age_dynamic" object.                                     |
| show_iqr         | Logical; whether to draw the interquartile band.                  |
| show_expectation | Logical; whether to draw the expectation curve.                   |
| main, xlab, ylab | Base graphics labels.                                             |
| add_legend       | Logical; whether to draw the plot legend.                         |
| ...              | Additional arguments passed to <a href="#">graphics::plot()</a> . |

**Value**

Invisibly returns x.

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